CHANDERNAGORECOLLEGE

B.A. / B.Sc. / B.COM (MAJOR/MINOR) SEMESTER – IV (CCFUP) INTERNALASSESSMENT EXAMINATION, 2025

SUBJECT: Computer Science COURSECODE:COMP 4012 COURSENAME:Programming in Java

TIME:30MINUTES

FULLMARKS:10

ATTEMPTANY01OUTOFTHEFOLLOWING03QUESTIONS

1x10=10

Q1: a) What are the differences between a concrete class, abstract class and an interface in Java? Which classcannot beinherited and which class cannot have instances?

b) Explain how you can create and manage threads using both the Thread class and the Runnable interface. Discuss the advantages and disadvantages of each approach and explain why using Runnable is oftenconsidered a better practice.

5+5

Q2: a) Differentiate between checked and unchecked exceptions, providing an example for each. Explain the purpose of the try, catch, finally, and throw keywords and how they work together to manage runtime errors.

b) Describe the key methods that demonstrate the mutable nature of StringBuffer and StringBuilder. Provide codesnippets for methods such as append(), insert(), and reverse(). Explain how these methods modify the object in place, contrasting this behavior with operations on the String class.

5+5

Q3: a) Describe the complete process of establishing a connection to a MySQL database using JDBC. Your answer should outline the key steps involved, from loading the JDBC driver to creating the Connection object. Include the necessary classes and methods you would use in a Java program.

b) Explain the core difference between method overloading and method overriding. Your answer should focusonthe concept of polymorphism—specifically, which type of polymorphism (compile-time or runtime) each featurerepresents. Discuss the differences between "Constructor Overloading" and "Method Overloading."

5+5